

Example 6: Let $a_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$, $a_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}$ and $b = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$. Then $\text{Span}\{a_1, a_2\}$ is a plane through the origin in $\mathbb{R}^3$. Is b in that plane?

Solution: Does the equation $x_1 a_1 + x_2 a_2 = b$ have a solution? To answer this, row reduce the augmented matrix $[a_1 \ a_2 \ b]$:

$$\begin{bmatrix} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 2 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3} - 3 \times \text{Row 1}} \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & -18 & 10 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ - 6 \times \text{Row 2}}]{\text{Row 3} = \text{Row 3} - 6 \times \text{Row 2}} \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

There exists a row in the augmented matrix of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. By Theorem 2, the equation $x_1 a_1 + x_2 a_2 = b$ has no solution, and so b is not in $\text{Span}\{a_1, a_2\}$. $\square$

Linear Combinations in Applications

$$\left\{ \begin{array}{l} \text{number} \\ \text{of units} \end{array} \right\} \cdot \left\{ \begin{array}{l} \text{cost} \\ \text{per unit} \end{array} \right\} = \left\{ \begin{array}{l} \text{total} \\ \text{cost} \end{array} \right\}$$

Example 7: A company manufactures two products. For \$1.00 worth of product B, the company spends \$.45 on materials, \$.25 on labor, and \$.15 on overhead. For \$1.00 worth of product C, the company spends \$.40 on materials, \$.30 on labor, and \$.15 on overhead.

$$\text{Let } b = \begin{bmatrix} .45 \\ .25 \\ .15 \end{bmatrix} \text{ and } c = \begin{bmatrix} .40 \\ .30 \\ .15 \end{bmatrix}$$

Then b and c represent the "costs per dollar of income" for the two products.